

Module 04 Lab - Exploratory Data Analysis

Objective: To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model.

This lab is fully coded. Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading

What is Exploratory Data Analysis (EDA)?

EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions.

We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns # Seaborn is a library built on top of Matplotlib
that makes creating beautiful plots easier.

# Load the dataset directly from a URL
df =
pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')

print("---- First 5 Rows ----")
print(df.head())

print("---- Basic Info ----")
# .info() is a great first command. It tells us the column names, how many
non-null values are in each column, and their data types.
# Notice that 'Age' and 'Cabin' have missing values!
df.info()
```

--- First 5 Rows ---

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

--- Basic Info ---

<class 'pandas.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	str
4	Sex	891 non-null	str
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	str
9	Fare	891 non-null	float64
10	Cabin	204 non-null	str
11	Embarked	889 non-null	str

dtypes: float64(2), int64(5), str(5)

memory usage: 83.7 KB

Part 2: Descriptive Statistics

Let's start by getting a high-level numerical summary of the data. The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
In [2]: # Get summary statistics for numerical columns
print("---- Descriptive Statistics ----")
print(df.describe())

print("---- Key Insights from Statistics ----")
print(f"The average age of a passenger was {df['Age'].mean():.1f} years.")
```

```
print(f"The overall survival rate was {df['Survived'].mean():.1%}.")
print(f"Fares ranged from ${df['Fare'].min()} to a whopping
${df['Fare'].max()}")
```

--- Descriptive Statistics ---

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

--- Key Insights from Statistics ---

The average age of a passenger was 29.7 years.

The overall survival rate was 38.4%.

Fares ranged from \$0.0 to a whopping \$512.3292.

Part 3: Visual EDA - Telling Stories with Plots

Numbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights.

A Note on Plotting Libraries:

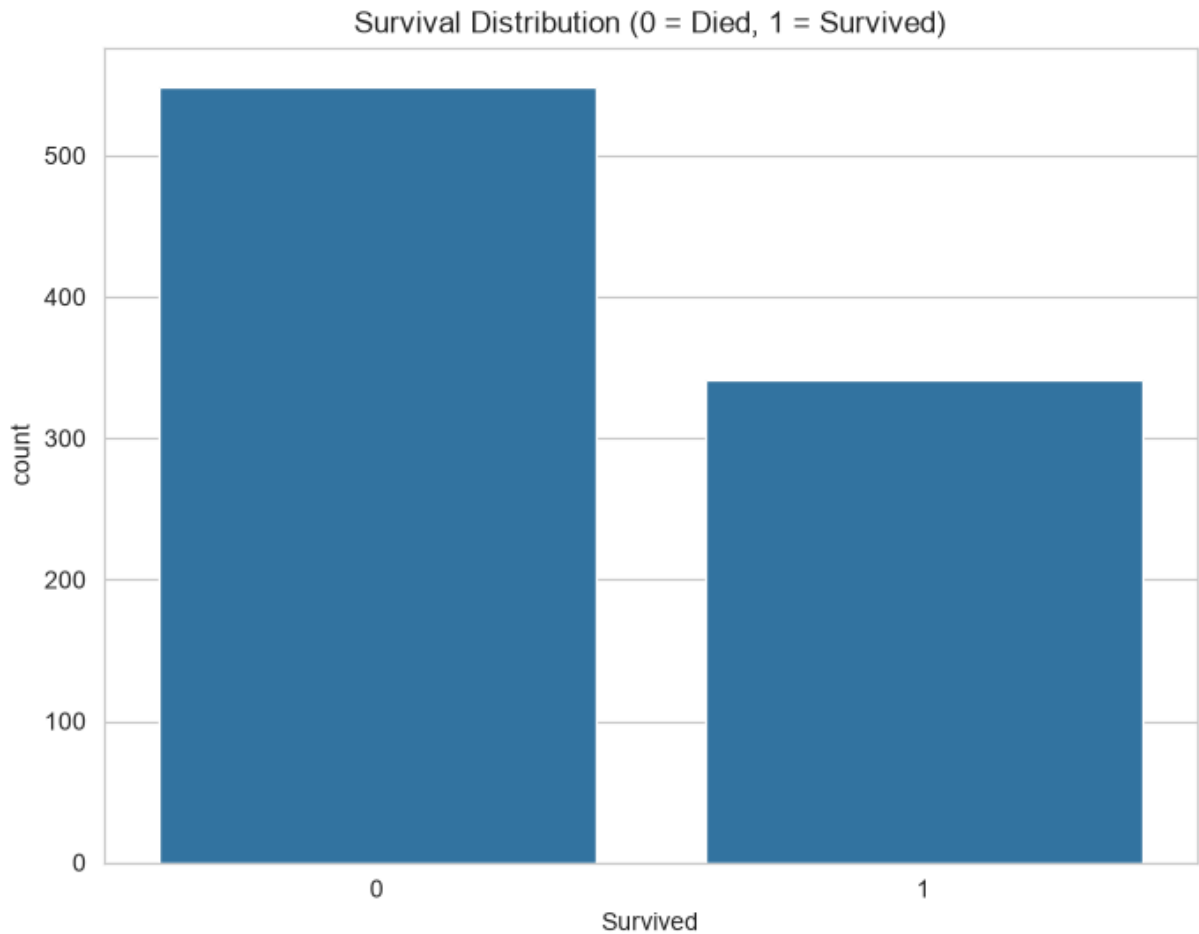
- **Matplotlib:** The foundational library, gives you full control over everything.
- **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

Visualization 1: How many survived?

A simple `countplot` is the best way to see the distribution of a categorical variable, like our target `Survived`.

```
In [3]: sns.set_style('whitegrid') # Sets a nice visual style for our plots
plt.figure(figsize=(8, 6))
sns.countplot(x='Survived', data=df)
plt.title('Survival Distribution (0 = Died, 1 = Survived)')
plt.show()
```

```
print("Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes be a challenge for machine learning models.")
```



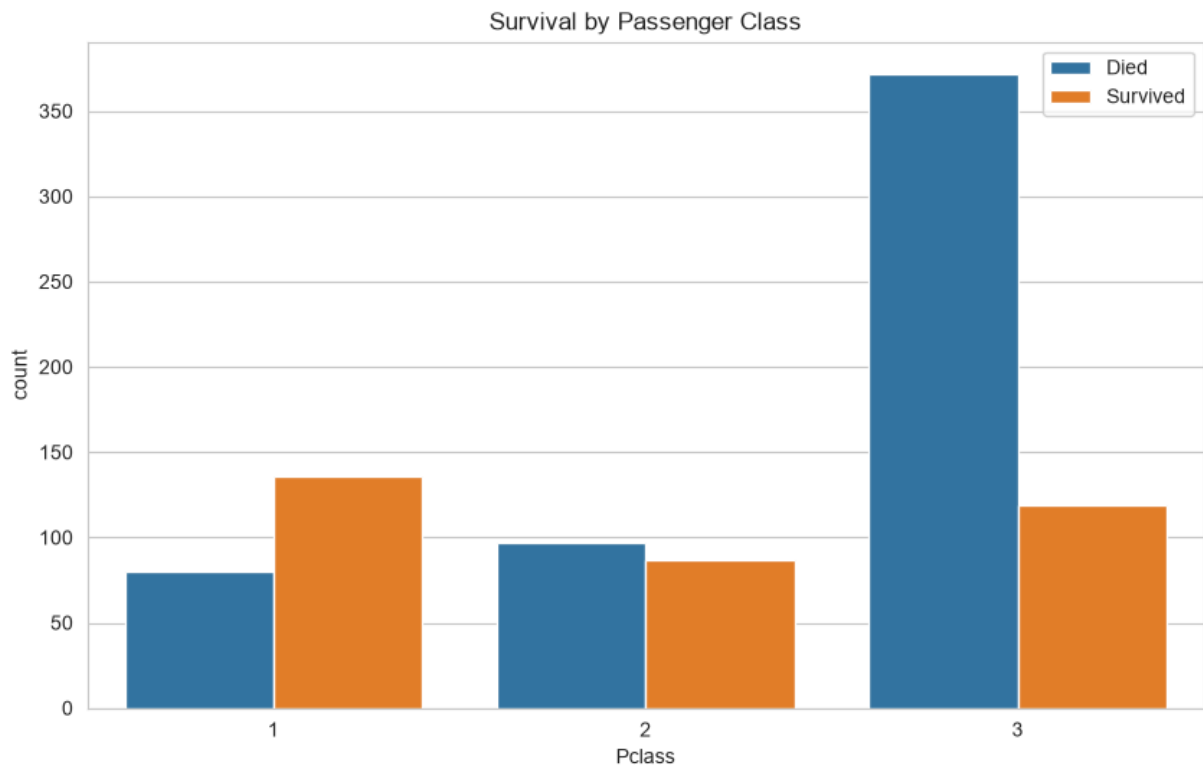
Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes be a challenge for machine learning models.

Visualization 2: Does passenger class matter for survival?

Now we want to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.

```
In [4]: plt.figure(figsize=(10, 6))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class')
plt.legend(['Died', 'Survived'])
plt.show()
```

```
print("Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class passengers. Money seems to have made a difference.")
```



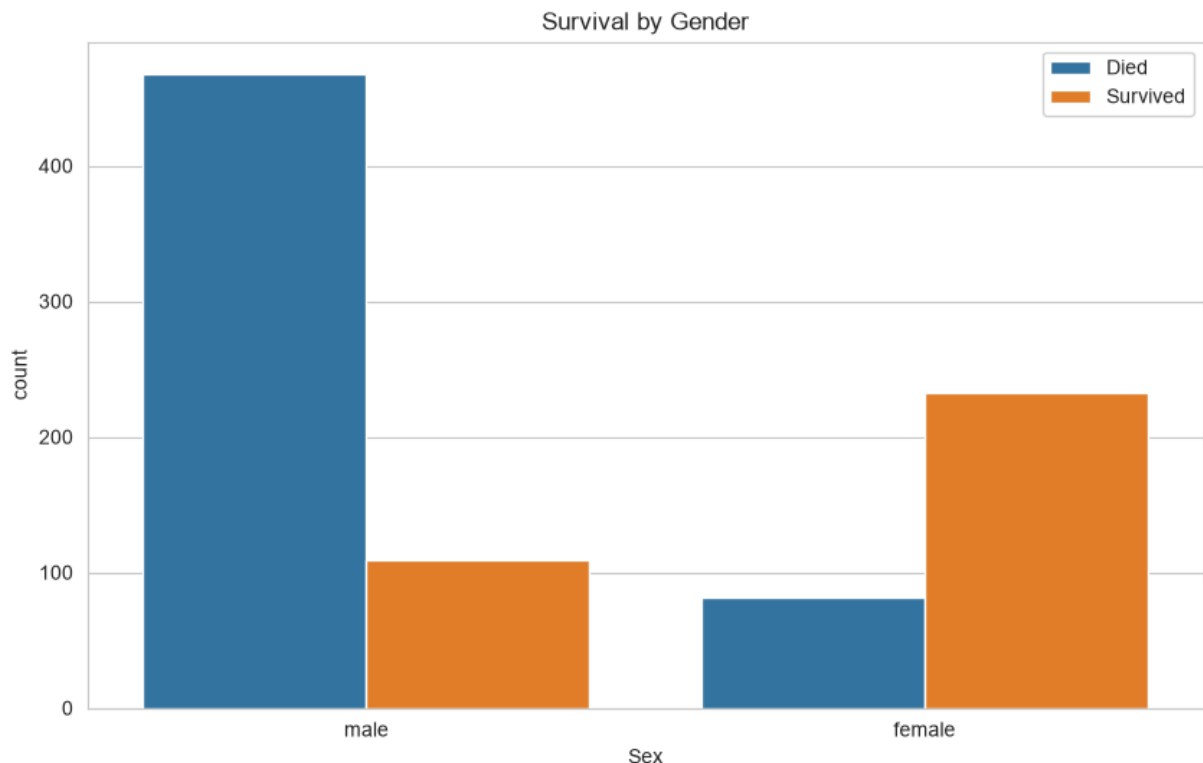
Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class passengers. Money seems to have made a difference.

Visualization 3: What about gender?

The 'women and children first' mantra is famous. Let's see if the data supports it.

```
In [5]: plt.figure(figsize=(10, 6))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.legend(['Died', 'Survived'])
plt.show()

print("Insight: The pattern is undeniable. A much higher proportion of
females survived compared to males. This is another very strong
predictor.")
```



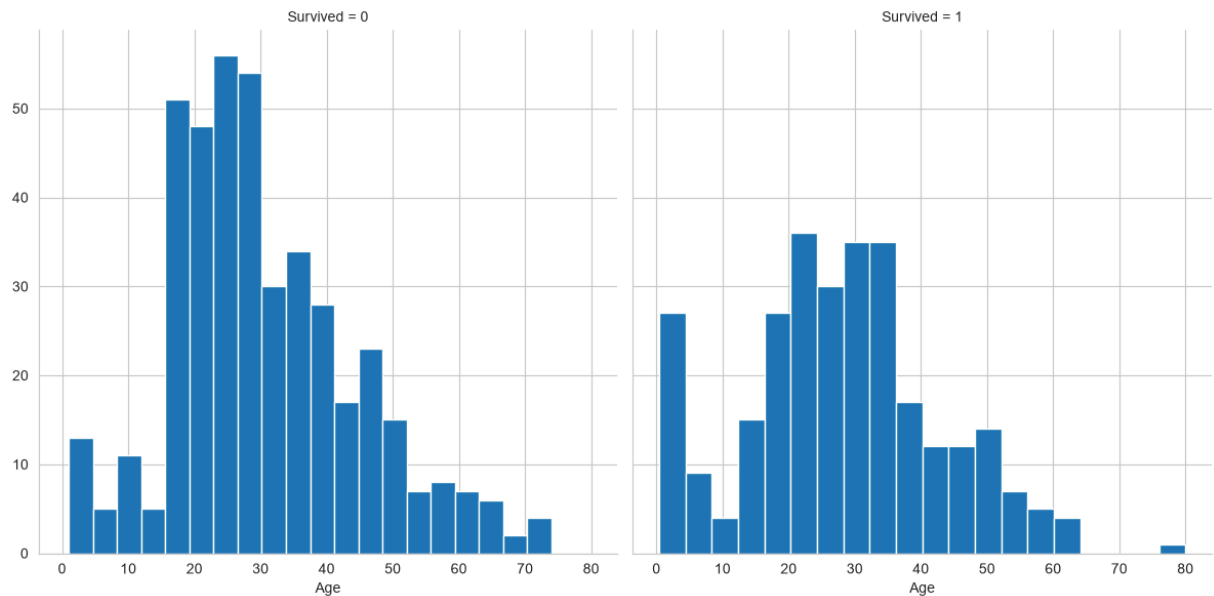
Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very strong predictor.

Visualization 4: How does age play a role?

For a continuous variable like `Age`, a histogram is a great way to see its distribution.

```
In [6]: # A FacetGrid allows us to create multiple plots side-by-side to compare
        # distributions.
        # Here, we create one histogram for passengers who died (col='Survived'=0)
        # and one for those who survived (col='Survived'=1).
        g = sns.FacetGrid(df, col='Survived', height=6)
        g.map(plt.hist, 'Age', bins=20)
        plt.show()

        print("Insight: The age distribution for those who did not survive is
        centered around the 20-40 age range. For those who survived, there is a
        noticeable spike for young children. This supports the 'children' part of
        the mantra.")
```



Insight: The age distribution for those who did not survive is centered around the 20–40 age range. For those who survived, there is a noticeable spike for young children. This supports the 'children' part of the mantra.

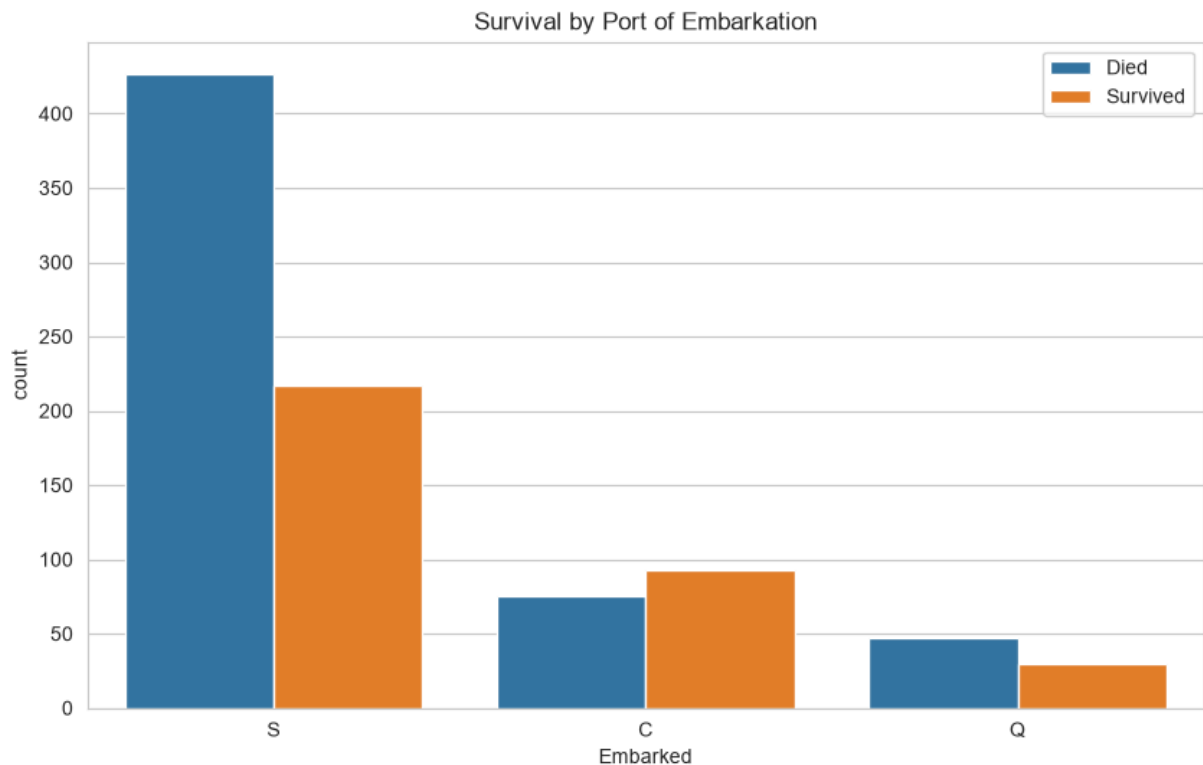
Part 4: Student Experimentation

Instructions: Create your own visualizations to explore other relationships in the data.

Experiment 1: Port of Embarkation

1. The `Embarked` column tells you where a passenger boarded the ship (C = Cherbourg, Q = Queenstown, S = Southampton).
2. Create a `countplot` to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

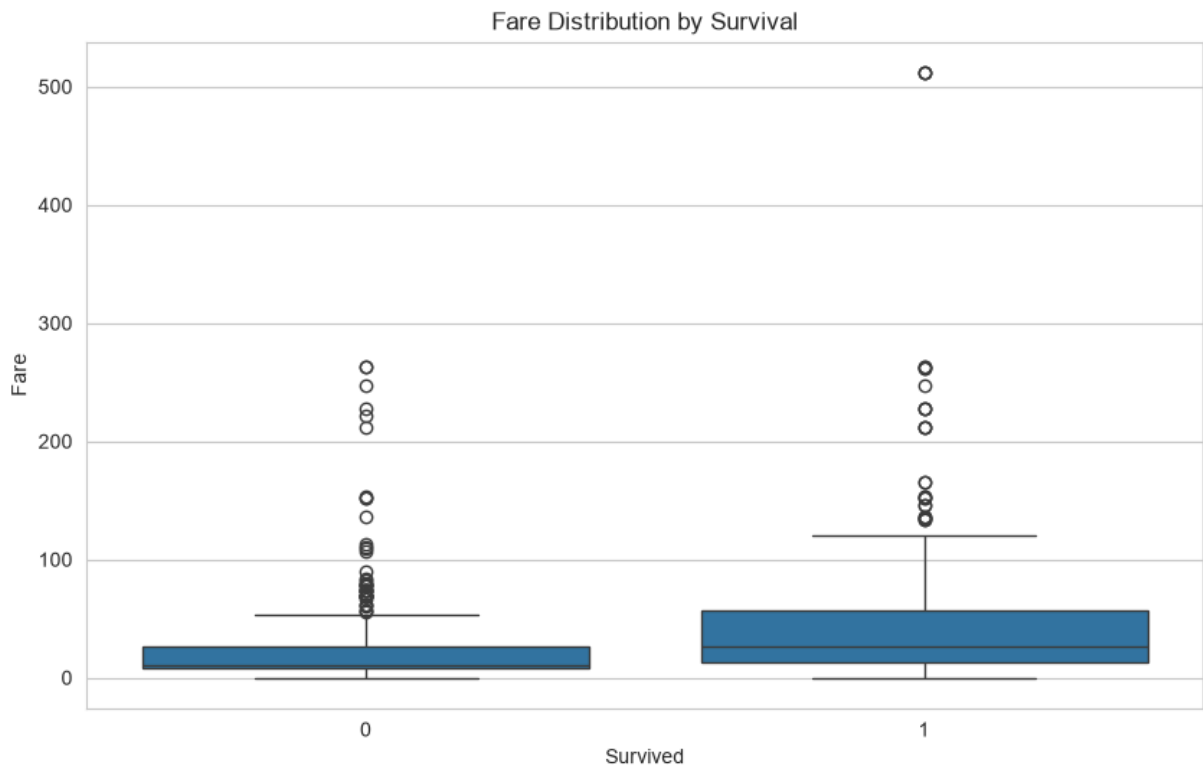
```
In [7]: plt.figure(figsize=(10, 6))
sns.countplot(x='Embarked', hue='Survived', data=df)
plt.title('Survival by Port of Embarkation')
plt.legend(['Died', 'Survived'])
plt.show()
```



Experiment 2: Fare vs. Survival

1. `Fare` is a continuous numerical variable.
2. A `boxplot` or `violinplot` is a great way to see the distribution of fares for those who survived vs. those who didn't.
3. Create one of these plots to compare the `Fare` distribution by `Survived`. What does it tell you?

```
In [8]: plt.figure(figsize=(10, 6))
sns.boxplot(x='Survived', y='Fare', data=df)
plt.title('Fare Distribution by Survival')
plt.show()
```



Knowledge Check

Instructions: Answer the following questions in this markdown cell.

1. **What is the primary goal of Exploratory Data Analysis (EDA)?**
2. **Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic?** (Describe them in terms of class, gender, and age).
3. **Why is it important to visualize data instead of just looking at summary statistics?** What can a plot show you that a number like 'mean' or 'count' can't?

Answer 1:

In my opinion, from what I've learned, the primary goal of exploratory data analysis is to essentially listen to the data and build up intuition before building any machine learning models. Essentially, EDA acts as the detective work phase. We're essentially inspecting all the data for structural discrepancies, trying to spot things like missing values or what they would call outliers. You know, we want to analyze and map future relationships and overall just verify that data quality. In doing this, this provides essentially, a blueprint for you know how we can subsequently clean and prepare data so the models that we're training it on actually get like the highest quality input.

Answer 2:

So yeah, based on my evaluation of the plots in this lab it looks like someone who was young, was considered first class, and a female generally had the overall highest chance

of survival.

Answer 3:

I feel like visualization of data gives you a more comprehensive picture instead of narrowing it down to statistical aspects like mean or count. Those don't show the entire data set. They're generally narrowed down to single numbers. And so, when you have more of a comprehensive scope of the data, you get to see more patterns. You get to see things like multiple peaks, which I learned is called bimodality, or things like extreme outliers. So the visualizations that were used here in the lab. I just feel show the entire shape of what the data was representing, revealing hidden patterns or clusters, even data eras that average numbers from a statistical standpoint might completely mask.